

The Illusion of Illusion

The Problem of Illusion is often taken as a paradigmatic objection to direct realism. Once the objection is admitted, the claim that we perceive reality without an intermediary appears to lose its hold, a host of further objections, including perceptual variation, become admissible, and we seem forced into accepting an explanatory posit along the lines of sense data. The aim in this essay is not to argue in favour of direct realism, but to show that, when understood correctly, the Problem of Illusion dissolves and that the world's disclosiveness, reality as that which reveals itself lawfully under determinate conditions, can account quite coherently for error and variation. In order to show this, I will explore claims about sense data and representationalism, as well as the ocular privileging given in standard accounts of perception. It will be argued that the standard objections to direct realism reveal themselves as a consequence of misconstruing of what reality is, that the paradigm is in fact a misdiagnosis and, once recognised as such, the illusory nature of the problem becomes apparent and the objection itself evaporates. In claiming this, my goal is to thereby sketch out the basis for an alternative theory of perception based on perceptual continuity and conceptually articulated constancy, which belongs to a larger, more unified view of the human.

The Problem of Illusion is considered to be an aporia for direct realism and a vindication of the indirect realist's account of perception. Direct realism's central claim that reality is perceived without an intermediary is thrown into jeopardy by the apparition of a thing which either does not exist or, via perception, differs from how it is 'in reality'. Direct realism's own solutions, such as we perceive sticks in water 'as' bent, appear insufficient partly because this kind of response seemingly concedes to a degree of validity in the indirect realist's accusation, giving the impression that, in order to survive, direct realism must defend weaker versions of itself, akin to the way the theist resubmits their definition of God after charges of incoherence. This attack by the indirect realist ultimately leads to the positing of sense-data and renders persuasive assertions such as, 'thus it becomes evident that the real table, if there is one, is not the same as what we immediately experience by sight or touch or hearing.'¹ While Russell, quoted here, is referring specifically to 'perceptual variation', an issue that will be addressed later, his point applies equally to cases of illusion proper, and his conclusion is one that explicitly contradicts the direct realist assertion that there exists between the perceiver and the perceived no intermediary. That is to say, only by positing an intermediary can illusion and perceptual variation be explained. That intermediary, for Russell, is sense-data or, more loosely, an 'appearance'.

The paradigmatic example of illusion is a stick bent in water, paradigmatic to the extent that it has become a cliché. Yet its ubiquity and apparent conclusiveness as an argument in favour of representationalism are here precisely what demonstrate its utility. Out of the water, the stick's shape is perceived as straight. Yet once in the water, a new perceptual

¹ Bertrand Russell, *The Problems of Philosophy*, Oxford University Press, p2

experience seems available to us: the stick now looks bent. Logic or reason confirms to us the reality of the situation: the stick acquires a distinct aspect whilst in the water under such conditions, without any material change taking place in the stick itself. The change is therefore, for the indirect realist, perceptual rather than ontological, pointing to the conclusion that there exists a gap between the perception of reality and reality itself.

If our perceptions differ from reality, reality is not perceived 'as it is', that is to say 'directly'. What, then, is being perceived? In Locke's case, *ideas* represent reality, or the perceived object. In the case of Ayer and Russell, sense data. While Russell and Ayer's views are not identical, they form a rough pair and it is with this understanding of perception, rather than Locke's specifically, that I will be concerned in this following section.

In *The Problems of Philosophy*, Bertrand Russell states that philosophy is oftentimes concerned with the difference between appearance and reality. The perceptual phenomenon of illusion is a case in point. The apparent bentness of the stick in water is not its reality, it is merely something that exists for the perceiver. In other words, perception of the world may or may not be how the world is, and there are times where it most avowedly isn't. What this position shares with Locke's is that of an assumed world-unknowableness. But unlike Descartes, for whom the certain lies within via clear, mathematical intuitions, the empiricist tradition to which Russell et al. belong has, in general, a mistrust of certainty and a somewhat tempered appraisal of the revelatory quality of all that is *a priori*. Consequently, 'knowledge' of the world must never be embraced too eagerly, and perception is regarded as fallible. Since a simple test, say, touching the stick, will reveal the stick's straightness, we are impelled to conclude that what we perceive is not necessarily the way things are but how they appear. Later, I will return to what I consider an inherent contradiction in this position. For now, let us follow the line of reasoning and explore exactly what is meant by Russell's term *sense data*.

For Russell, visual perception and touch give us information about the world which is of doubtful certainty because it is inherently subject to variation. The colour, texture and shape of a table are all liable to change depending on the position of the observer, the time of day, the amount of light available in the room and so on. Let us focus on Russell's observations as they pertain to the table's shape. The table's 'real' shape, he argues, is something not directly available to us, but something we construct through practice² - experience, akin perhaps to how, for Hume, we begin to 'perceive' causality through association and make mental connections. For Russell, we 'construct' the table's 'real shape', for example, a rectangle, then interpret what is before us, rather than perceive it directly:

² *ibid*, p3

The 'real' shape is not what we see; it is something inferred from what we see... so that here again the senses seem not to give us the truth about the table, but only about the appearance of the table'.³

For Russell, this 'appearance' is the overall perceptual gestalt. What comprises this gestalt are what he terms 'sense-data', i.e., those things that are 'immediately known in sensation'⁴, namely shape, colour, texture, size, and so on. The mahogany red of the table is a 'sense-datum', we are aware of it, and in becoming so, we have a sensation. Awareness of sense-data is sensation. Sensation is the experiential awareness of such immediate sensory particulars. Our engagement with reality, then, is mediated: it takes place via appearance, comprised of sense-data elements, and the underlying or 'beyond' object is represented to us, constructed by our experience. The mind organises and interprets sense-data, so the object as we know it is partly a construction from our sensory input and cognitive processing. In short, on this view, the subject constructs the representation of the object.

This line of thinking leads to Russell's saying that 'the table, if it exists, we will call a physical object'. Reality is, if knowable at all, known indirectly, doubtfully. The sceptical tone which emerges here is no accident. Indeed, one of Russell's parting comments of the chapter is that 'the one thing we know about [the table] is that it is not what it seems'⁵. It is a series of appearances comprised of sense-data; probably real, but always out of reach. Being the direct objects of awareness, these sense-data are private and subjective. While Russell is no solipsist, his position does commit us to a theory of perception which places an epistemic shield between the perceiver and the perceived.

The gains of positing sense-data are most in evidence when we consider the issues of perceptual variation and the case of illusion. Given that it is appearances we perceive, the changing colour and shape of the table becomes explicable: the table itself does not change but the appearances vary, relative to one's vantage point, the bentness of the stick is ocular rather than structural. Appearances account for variations and tricks of light show that we sometimes see what is not necessarily the case. As mentioned, the direct realist may reply that we perceive the stick as bent, preserving the claim that we perceive reality directly. Initially, this seems to be working in service of the indirect realist, so much so as to be barely distinguishable from what sense-data theories themselves claim, i.e., that while the object is 'one way', our perception of it is 'other', albeit without employing the term 'appearance'. Later, I will argue that this alleged own goal, properly understood, does in fact contain the germ of some kind of solution to the problem. For now, let it be said sense-data make illusions explicable, and that illusions make the positing of sense-data cogent. By explaining illusion thus, a number of things are thought to be achieved. We are often mistaken about what we perceive in the world, as Descartes lamented. The

³ *ibid*, p3

⁴ *ibid*, p4

⁵ *ibid*, p6

fallibility of the senses reveals itself in our mistaken judgements, in believing a thing is one way when 'in reality' it is another, perception is filtered through our sensory and cognitive apparatus, thereby accounting for the subjectivity of experience, the unknowableness of the world becomes tolerable, illusion has been made intelligible, and even if it can't be ignored or 'not perceived', it can be understood. This explanatory recourse to appearances is a prudent, grounded position which inoculates itself from dubious metaphysical assumptions by reminding us to retain a cautious, Lockian humility about the limits of knowledge. And while we may never perceive the world as it 'really' is, we can at least be aware that illusions exist.

But before we begin to assess the efficacy of this solution, it would be well to acknowledge some of the costs of accepting this view. While it appears to have solved the issue of how things may be one way yet seem another, we appear to be committed to a degree of scepticism, as Russell's 'the table, if it in fact exists', points to. The mystery has gone: reality filters through the subject and may or may not accurately represent the external world. This explains illusions and perceptual variation, but commits the sense-data theorist to accepting the view that the world itself is ultimately unknowable. An even worse scenario is that the external world simply does not exist, but Russell discounts this. It remains for him an 'uncomfortable possibility', but this 'possibility' in fact only arises once we accept Russell's reasoning in general. And while he discounts the most sceptical conclusion, his solution to appearance versus reality posited by sense-data leaves him beset with uncertainties. That is to say, the alleged unreliability of the senses appears to solve one problem while generating numerous others. Rather than argue, however, whether scepticism is simply the price to be paid for a convincing theory of perception, it might be more profitable to ask if Russell's theory is in fact convincing.

Illusion is taken as decisive: there are occasions when we do not perceive 'what is', but 'what is not'. The minimum claim, then, is that on occasion we are not perceiving 'reality'. To explain this, it is argued that we perceive, at least on these occasions, a representation or appearance, which accounts for the disparity between how things really are and our perceptions of them. The element interposed between the world and us is sense-data. Yet it would be odd for sense-data to arise only when illusion type cases proliferated - why would they intervene at such moments? What would provoke their apparition? How might we, or our minds, distinguish, even unconsciously, between moments of non-veridical perception and veridical, so as to produce relevant sense-data for the given occasion? It seems more likely that, if sense-data exist, they do so always, rather than occasionally. Therefore, illusion, in accounting for the disparity between perceptual appearance and reality via sense-data, defeats direct realist theories of perception and safely establishes a theory of perception which argues that we perceive the world indirectly, representationally, through a mediator. Representation stands between us and the world, with all the sceptical implications that this entails. However, a promising objection to this can be offered by exploring the phenomenon of refraction.

Refraction is the phenomenon of a change of direction in a wave, typically of light, brought about by its passing from one medium to another. When light enters water the speed at which it travels decreases. When the light hits such a surface at an angle one side of the wave slows first, causing the entire wave to turn; what we describe as 'bent' is the result of this change in direction, or 'pivoting'. The straight stick, half submerged, looks bent, the light from submerged part seeming to incline away from the normal as it leaves the water. For the indirect realist, the move is decisive: the stick is known to be 'straight', yet now looks bent, therefore, we clearly have a case of reality appearing other than it is, hence perception deals not in direct reality, but in appearances. Direct realism itself here offers little in the way of a credible refutation. One response open to it is to acknowledge that while we perceive the stick as bent, we understand that 'in reality' it isn't. Even as children we cease to be fooled by what we have seen once an adult has explained to us 'the truth' of the situation. After that, each time we encounter refraction, our mind corrects the visual perception and we come away with a secure sense of reality all the same. If then asked to explain what it is we are perceiving, the direct realist is entitled to say that, in that moment, we perceive the stick as bent, the stick appears bent and we perceive that, rather than an appearance. Sense data remains, then, an unjustified move. But this reply is hardly satisfactory. For one thing, using the terms 'appears' and 'appearance' feels self-defeating, allowing into the defence the language of the interlocutor. The claim that we perceive how a stick appears while knowing the stick to be otherwise differs little from the claim that we perceive appearances rather than things in themselves. Worse, the defence itself offers almost no explanatory content whatsoever. It does not, for example, account for why refraction should show itself to us in this way, nor does it strengthen the claim that perception consists only of a perceiver and a perceived. Rather, it seems almost to accept the terms of the objection and to try to pass them off as being congruent with its own position. So far, then, illusion, especially in the case of refraction, seems to remain a decisive attack against the notion that we perceive reality without mediation. But is the above defence as simplistic as it seems?

Austin writes of the bent stick, 'we can see that it's partly submerged in water, so that is exactly how we should expect it to look.'⁶ Familiarity, he claims, 'takes the edge off' illusion⁷. He is attempting to clarify, as per his ordinary language project, what counts as illusion proper - and unfamiliarity must therefore be one of illusion's necessary conditions. But, in 'taking the edge off' in this way, he seems to feel he has neutralised the issue. Several pages later, he makes a similar point in relation to Ayer's claim that 'at least one of the visual appearances of the stick is delusive'⁸, asking if the stick truly looks bent, concluding that, 'it does not look exactly like a bent stick', but rather like a 'bent stick partly immersed in water'.⁹ He thus questions Ayer's use of 'delusive': exactly how is it

⁶J. L Austin, *Sense and Sensibilia*, Oxford University Press, p26

⁷ *ibid*, p26

⁸ *ibid*, p29

⁹*ibid*, p29

that we are meant to be deluded? Austin seems, to be concerned to dissolve the issue by saying none of that which we perceive in this instance should in any way be a source of surprise to us, and rightly so. The problem, however, lies in his supposing that by simply raising these rhetorical issues, he has done more than merely point out that the sense-data-appearance problem has been misconceived, that illusion is an illusion. What Austin's line of defence does not supply in any substantial way is an explanation of what in fact is happening, or of how reality itself operates. He writes:

...what we see is a *stick partly immersed in water*; and it is particularly extraordinary that this should appear to be called in question... since this, after all, is simply the description of the situation with which we started. [*ibid*, p31]

And this is where I find his observations to be of limited use, even if they might serve as a point of departure. In fairness to Austin, we can say that he achieves his purpose and that charging him with failing to do things he never intended in the first place is uncharitable. Austin himself is involved in an enterprise of linguistic hygiene, the fulfilment of which is expected to bring epistemic clarity and comfort. Let us try then to extract from his critique an insight which may well contribute to an ontology of perception.

Leaving aside Austin's ordinary language philosophy aims, the key insight worth taking as a point of departure, and which aligns with the generalised direct realist response critiqued above, is that we not seeing an illusion so much as the reality of a stick in water. On the surface, this response seems to do no work whatsoever, nor does saying that the paradigmatic case does not count and therefore sense-data becomes an unjustifiable hypothesis aimed at explaining something which is already comprehensible. The explanation is clear: we are witnessing a case of 'refraction', and given that we know how refraction works, such 'knowing' reveals there is no illusion. The issue is that refraction is taken for granted, and what it could potentially reveal about perception is being overlooked. It is to these potential revelations we must now turn.

Austin's claim that seeing refraction leads us to see precisely what we should expect points to an important consideration. Refraction qua phenomenon is: predictable, common and repeatable under very precise conditions. That is to say, it would be surprising if, given said conditions, refraction didn't take place, i.e., if the stick did not bend. Ayer's assumption is that refraction is a thing which conceals reality from us, with the truth only reaching us via the distortion of appearances that we are able, post-factum, to mentally ascertain. The suggestion is that because a partially submerged stick is perceived as bent, the revelation this affords concerns our perceptual limitations, plus the concomitant existence of a mediator between us and the external world. In fact, it reveals no such thing. The ubiquitousness of this phenomenon under said conditions, its predictability, reveals something quite the opposite. First of all, that refraction is a law-like phenomenon that obeys the rules of physics. Secondly, that our perceptual faculties disclose this phenomenon faithfully. Thirdly, that reality itself is disclosive. Taken together, these issues point to a very different dynamic between perceiver and perceived and the way in which we, as beings capable of perception, are in the world in relation to it.

The first point has been stated, but let it be added and further clarified that refraction is in no sense arbitrary. It is, in addition to being predictable, quantifiable and law-governed. The occurrence of a 'bent stick' is therefore neither a random distortion nor a subjective illusion, but a mode of revealing grounded in the structure of reality itself. Earlier, it was stated that:

1. The stick looks bent
2. Yet it is really straight
3. Therefore, perception gives us appearances, not reality

This can be inverted:

1. The stick looks bent
2. Physics explains exactly why it must do so
3. Therefore, perception is tracking the real behaviour of light in the world

This is Austin's 'precisely what we should expect' with its fuller ontological implications starting to be teased out. The so-called 'illusion' points to the fact that perception is embedded in reality, not detached from it.

The predictable, repeatable nature of refraction, the fact that such 'illusions' (specifically of this kind), do not appear outside of these situations, in unexpected or inexplicable contexts, reveals something more along the lines of the *reliableness of the senses*, which we might better call 'the lawfulness of perceptual disclosure'. For Descartes, the unreliability of the senses serves a methodological purpose: perception is called into doubt in order to secure certainty elsewhere, namely in the clear and distinct deliverances of thought. The instability of perception is thus acknowledged, but not yet decisive. In Locke, this balance begins to shift. While the existence of an external world is retained, what is immediately given are ideas, and the thing itself becomes something only indirectly accessible. The possibility arises that the true nature of the world may never be fully known. Berkeley eliminates this gap by denying material substance altogether: what is given in perception is all that exists. Yet this move comes at the cost of collapsing the distinction between appearance and reality rather than resolving it. With Hume, the consequences deepen further. Not only is access to the world mediated, but even the structures we take to belong to it, such as causation, are revealed as products of habit rather than features of reality itself. By the time we reach Russell, this trajectory culminates in a fully articulated separation: what is immediately given are sense-data, while the object itself becomes an inferred posit, 'the table, if there is one'. What begins as a methodological doubt in Descartes culminates in a structural division between perceiver and world, in which perception is no longer taken to disclose reality but to stand in need of correction. And yet, refraction offers us a concrete point of contact not only with how the world 'looks', but how it functions; the distortion we perceive is not to be judged an imprisoning limitation of the human mind, but a disclosure of the physical laws of the universe.

If perception is reliable, a number of things follow. Refraction takes place, distorting how a stick would otherwise look out of water. But instead of mis-perceiving the stick in water, we register it as it is under such conditions. It is not 'not bent but perceived as bent', it is 'distorted by conditions', and it is *those distortions themselves* which are perceived. If our senses were unreliable, it is arguably the case that we would see the stick as straight when *in* water, which would itself be a distortion. It follows that perception enables us to participate in the unfolding of reality. The activity by which we make sense of how the world behaves is the intellect itself articulating what perception discloses; in the sense that Austin says we 'know the stick isn't really bent', we then correct our understanding of what is perceived via a reasoning processes in view of the fact that the perceived does not tally with generalised experience, i.e., sticks tend not to be bent. The reliability of perception then leads to a discovery: under specified conditions, a stick looks bent and this in turn increases our understanding of reality: light behaves this way, and what follows for perception is that, offering partial disclosure, it combines with intellect, achieving an integration of disclosure. Unreflective perception might misguidedly merit the epithet 'unreliable', but perception is better characterised as one element within a unitary process, in which intellect judges what perception discloses. Surely, when Descartes judges the men in the streets to be sticks with hats on, or when Russell is unable to determine the 'correct' angle of the table, the problem lies not with what is given by perception but in the conclusions that are reached? Surely if we hear a dog bark but conclude it is the burst of an exhaust, that is heard is determinate either way. The view we form as to what we heard cannot change what was disclosed in hearing, even if we can reconfigure its significance. To the trained ear, Schoenberg sounds like a breakthrough in music while to some a mental breakdown; yet the *sound* remains the same. In like fashion, refraction to the indirect realist is proof of appearance mediating reality, of the unreliableness of the senses and the ultimate unknowableness of reality. And yet, understood as a unified process in which what is given in perception is taken up and articulated, perception becomes a moment in a participatory exchange between perceiver and world wherein the way reality is avails itself to the perceiver. We can say that the two aspects of this process are: perception giving what is there under conditions, with judgement giving what it is taken to be. When Aristotle writes that 'thus it is for a man to think, whenever he will, but not so for him to perceive, because for that the presence of a sense-object is necessary',¹⁰ he is saying that perception is constrained in a way that thought it is not because it requires a sense object. Both Descartes and Locke make a similar point, albeit in order to defend the notion of an 'external world'. But where these thinkers run into trouble is in naming perception the fallible element, while 'thinking' is the corrective part. It seems more likely, however, that the reverse is true. Perception discloses what is there under conditions; the intellect determines what it is taken to be. Error belongs to the latter, not the former. Misinterpretation takes place at the level of thought rather than perception because perception itself is not interpretative; since, as Aristotle says, it is tethered to reality - by the necessary presence of a sense-object. The intellect can 'operate at will'. Seen in this light, perception and thought are not two

¹⁰Aristotle, *De Anima* II.5, Penguin, p171

separate activities but two elements within a single engagement with the world, perception initiated *by* the world and thinking responding *to* and working *on* what is given by the world via perception. Error is when what is given in perception is taken to be something it is not, because the intellect misjudges what perception discloses. Thus, perception and thought are distinct in kind but continuous in activity.

That is not to say that reality is always fully availed; some animals have fewer senses than others, some humans lack sight or hearing, but this does not affect how the laws of physics are constituted or operate, and light refracts in water whether moles, human, both or neither exist. And if refraction is both predictable and perceivable, i.e., the submerged bent stick enables us to perceive that refraction is taking place - and how else would we have discovered this law of physics - sense-data no longer occupies an explanatory place in the chain of events, but rather comes to seem like an unwarranted, indeed, unnecessary, postulate.

It might be claimed that even if the case of illusion can be reversed, especially in virtue of its comparatively limited or local occurrence, the above observations will exert less pressure on the more vexing, and pervasive, problem of perceptual variation. Russell believes that the impossibility of finding a correct vantage point from which to view an object is further proof that perception deals in appearances first, reality second - if at all. He posits a specific table and accords it the following attributes: oblong, brown, shiny, smooth, cool and hard, a description with which, he says, anyone who perceives the table 'will agree... so that it might seem as if no difficulty will arise'¹¹. But a difficulty does arise when we consider the table's colour. For while Russell believes the table is 'really' the same colour all over, the parts that reflect the light look much brighter than the other parts, and some parts look white because of reflected light. Thus, while we judge the table brown, it becomes impossible to locate the correct position from which to make the judgement, since each position offers up its own variation. Indeed, '[the table] appears to be of different colours from different points of view, and there is no reason for regarding some of these as more really its colour than others, such that 'we are compelled to deny that, in itself, the table has any one particular colour'.¹² Extending this to the concern of the table's shape allows Russell to make his decisive move. While we 'know' the table to be rectangular, this is something we in fact never see but which experience has 'taught us to construct', *ibid*, p2, for what we see are endless variations of converging lines, acute and obtuse angles, parallels pointing away from the spectator, very much in the manner of a perspective drawing. Russell is adamant about what is taking place: such variations and the impossibility of finding a neutral or objective vantage point in which 'the rectangle' is visible without distortion, to which we might return to achieve epistemic respite before returning to the kaleidoscope of unreliable angles and points of view, is an impossibility. Such a zero-point of perceptual clarity does not exist; thus, 'the table' itself

¹¹ Bertrand Russell, *The Problems of Philosophy*, Oxford University Press, p2

¹² *ibid*, p3

is never perceived or experienced; rather, 'the table', its reality, is inferred from what the impoverished data which perception yields in the form appearances:

The 'real' shape is not what we see; it is something inferred from what we see. And what we see is constantly changing in shape as we move about the room; so that here again the senses seem not to give us the truth about the table itself, but only about the appearance of the table.¹³

Given that perceptual variation is an ineluctable and omnipresent feature of perceptual experience, and refraction is by comparison far less common, perceptual variation may count as a species of illusion or serve the same critical purpose, and serve it more powerfully. Russell is surely right to say that the 'rectangle' remains elusive. But this elusiveness does not show that we fail to perceive the object. Rather, the variability of appearance belongs to the object's mode of disclosure; our concepts stabilise this variability without generating it. With this in view, we may ask: what is a rectangle? What constitutes a 'correct vantage point'?

A rectangle is a stable structure disclosed across a range of perceptual variations. What we call its 'definition' is the articulation of this structure, not its source. As such, it is distinguishable from a square, a circle, or any other kind of shape. From this agreed definition, it is easy to pick out examples amongst the items around us: desks, phones, books. True, whenever we encounter such an item, we run into the difficulties Russell is drawing attention to - but not necessarily in the way he says, nor even leading to the conclusions he reaches. A book on a table will likely be at an angle when we encounter it, so the acute and obtuse angles he evokes are things we perceive. This, for Russell, invalidates the notion of a 'real rectangle', but notice a peculiarity of his claim: the notion of 'real acute angles' is not denied but in fact invoked. If rectangles were impossible to perceive directly but only as appearances because we see instead only acute or obtuse angles, surely it follows that we would be seeing appearances of those angles themselves? Moreover, while an object may appear to us at an angle, it does so in a manner congruent to itself, which is to say, the definitional articulation we have given to that particular shape in response to the structure it discloses- a 'rectangle' seen at an angle is understood as being a rectangle and as behaving as a rectangle does, not say, as a sphere behaves. A square, seen at a similarly 'sub-optimal' vantage point, is not likely to be confused with a rectangle, or a triangle. The variations Russell alludes to are not arbitrary, nor is there any reason to call them subjective. The same 'distorted' angles are repeatable and predictable, and in this way they recall the phenomenon of refraction. A triangular piece of card lying flat on a table seen at X distance is identifiable as triangular, not inevitably to be confused with another shape. Russell seems to want to be saying that a triangle only truly appears as a triangle if perceived at a certain optimal height, distance, and so on, without 'distortion'. While it is true that this is how we tend to imagine triangles, a triangular-shaped object lying on a table behaves in a certain way proportionate to it what it is, recognisable as such.

¹³ *ibid*, p3

The sense in which a triangle, or any shape, is what it is lies in the fact that we have agreed on the necessary and sufficient conditions that articulate a stable and recognisable structure inherent in the thing itself. This is as true for Russell's table as it is for its shape. But it isn't the case that we claim that, due to perceptual variation, we sometimes perceive a table while at others worry we are perceiving a hippopotamus. The tableness, contained in its function, is less in question than its shape, which refers to rectangles. True, Russell allows himself to doubt that there is a table, but this is part of the sceptical concern that striving for a 'correct vantage point' generates. Let us summarise as follows:

A rectangle is two things. 1. a determinate geometric entity whose definitional account has been agreed on in accordance with a stable and recognisable structure inherent in the thing itself. 2. an abstract concept which finds its particular, substantial instantiation in a given physical object, e.g., a table, an envelope, etc., which behaves in quantifiable, predictable ways in readily replicable conditions, and as such is identifiable qua rectangle via, and not in spite of, perceptual variations.

As refraction pointed to a disclosure of physical laws which, if properly understood, contribute to the enlargement of human knowledge, so too do such variations point to predictable geometric structures - and to a deeper ontological issue. The identifiableness of a rectangle via its predictable variations points instead to *perceptual constancy*. Russell's error is to presume that 'rectangle' names a single component, the continual inaccessibility of which leads him to conclude that what is perceived is appearances of things, such that the object is reconstructed once experience has taught us how. This approach fundamentally misunderstands how perception functions. 'Rectangle' isn't singular per se but in fact a coherent, finite system of possibilities, all of which refer back to and meet the necessary and sufficient conditions of how rectangle is defined in relation to the recognisable structure that it is. Thus, reality itself comprises multiple integrated elements. Concepts are agreed on and give name to mind-independent objects, which in turn are identifiable through and not in spite of variations. This is not conventionalism, nor a Kantian shaping of reality via the mental. Rather, human existence is both physical in the world and abstract in organisation of concepts. Objects' having certain properties are picked out through perception, reflected on and given definitions which then serve as a basis of communal intelligibility. We might term this 'conceptually articulated constancy'. What variations reveal is not that we perceive appearances at all but that the concepts we use to describe reality, and language itself, are/is limited: 'rectangular' refers to one kind of shape regardless of whether the shape is perceived directly ahead or at an angle. A tilted rectangle is still a rectangle.

Mind-independence, then, is presence of stuff, and mind-dependence is the human schema used to describe and define that stuff encountered in the world via perception. Thus, when variations occur, either in the case of the table and shapes, or in the case of refraction, nothing in the world actually contradicts itself; rather, the world is behaving according to predictable physical laws. What is revealed for the human is our constantly

improving efforts to name what is taking place. Perceptual shifts are a revelation of a thing's being and predicates such as rectangular are tools of discourse. The world's reality isn't a single frozen view but rather the total field of possible perspectives that belong to the same material presence.

The second question, that which asks 'what is the correct vantage point?' now becomes answerable. Russell's mistake, and Berkeley's when he speaks about clouds, is to suppose that there must be a single, privileged perception point/vantage point. For them, the absence of such exposes the failure of realism.

Russell asks what is the correct colour of the table or position to view it from in order to correctly perceive its shape? But framing it this way almost becomes rhetorical. The standard, correctness, can't be met, and an erroneous conclusion is reached. The error lies in taking *correctness* as the valid criterion in the first place. Since it cannot be met, the conclusion seems inevitable: there is no such thing as 'correctness' here. And so by imposing the wrong criteria a false problem is generated, constituting an illicit move that Russell smuggles in. By suggesting that 'singularity' equals correctness, and by showing that no singular vantage point does in fact exist, Russell moves to the conclusion that subjectivity and appearances must be the stuff of perception. But the problem has been misconstrued, and when we can see that, Russell's position collapses. To answer the question of what constitutes 'correctness' in this case, we should keep in mind Aristotle's dictum that we demand only as much precision as is feasibly to be expected from a given field, and when we do that, the answer appears, namely, what is required in the case of perception is an appropriateness relative to disclosure - sufficient for the revealing of what a thing is.

Modifying Berkeley's example about clouds, i.e., what the 'true size' of a cloud is given that it appears small from a distance and larger nearby, we can explore the following: the framing of 'how close do we need to be to see the real size of a flower' is a misleading question which in effect engineers an anti-realist or sceptical outcome. The size is determined by agreement in relation to structures inherent in the thing itself. If the size is two feet, this is agreed on and acknowledged in advance and is not subject to charge, regardless of perceptual experience. In which case, in order to ascertain the 'agreed on size' of the flower, via measurement, the 'correct vantage point' is whatever is sufficient for measurement to take place, i.e., presumably a space containing at least two feet itself. This measurement does not change and is as fixed as the term 'rectangle'. The more revealing consideration is, for *what* is the perception taking place? The requirements of a florist and a botanist are different. Under a microscope, in which the flower appears granular or rock like, the botanist is able to work while the florist does not recognise what they are looking at. The correct distance to view the flower, then, is relative to the purpose of the perceiver. For the botanist, the microscopic view is the optimal condition under which to view the flower; for the florist, several feet back allows them to place the flower in an arrangement. Recalling Aristotle again, the judgement as to what constitutes 'correct' is relative to one's predicament, just as was Milo of Croton's protein intake. In

both cases, the conceptually articulated constancy of two feet, both something within the structure itself and something given linguistic expression by us, remains throughout. One objection that suggests itself here is that these two examples concern situations which are 'purposeful', and that perception in general is not consciously purposeful, nor is that what Russell is getting at. What would be, then, a relative vantage point for perceiving a flower? The issue which all these considerations are circling is one of *ontological optimacy*. The correct vantage point, or points, for perceiving a flower in a field are those which facilitate the disclosure of the flower qua flower. Too far or too close and the flower does not reveal itself, but at a certain distance what it is is availed to the perceiver. Meanwhile, the fixed, agreed upon measurement of two feet which is an articulation of structures inherent in the flower itself remains constant throughout. 'Optimal distance', therefore, is not epistemic privilege but ontological fit. This understanding of perception echoes Merleau-Ponty's reference to a painting in a gallery. When the perceiver is in such a relation to the perceived that ontological disclosure of what a thing is can take place, the position is optimal. This may be either purposeful or accidental, and the majority of cases of perceptual experiences very likely have little that is inherently purposive about them. But the relative mean and context-dependence drive the definition of 'correct' (optimal), denying an absolutism which would force us into a sceptical retreat, and the conceptually articulated constancy, in this case measurements of feet, shows both the utility and limits of our attempts to conceptualise the world. By replacing 'Absolute correct distance' with *appropriateness relative to disclosure* such that the revealing of what a thing is can take place, we can see that variation is no more of an objection than refraction. Variation in fact points to modes of disclosure, each with their own relevant set of criteria. Depending on our relationship to, intention towards, position to, the perceived, certain elements of the thing will be more accessible or highlighted while others will remain, in that moment, undisclosed, though always with the potentiality to be disclosed. Thus, perceptual variation does not threaten objectivity, it only shows that intelligibility is embodied and situated. Moreover, replacing 'correct' with 'optimal' is not a mere linguistic fix, but an ontological shift. It relates not to a supposed 'correct' measurement or distance, or time of day, but to the being of the thing perceived, the what it is, so that this what it is can be disclosed in its ontological fullness.

By misdiagnosing the problem, i.e., that 'correctness' is required but is impossible, Russell is forced into positing various metaphysical claims such as sense-data, appearances, the unknowableness of reality and even the latter's possible non-existence. These conclusions are avoidable. When perception is taken as essentially representational, an act of reconstruction based on what experience has taught us, perception becomes a bridge from subject to object; this kind of lineage, for all it strives not to be, is as fundamentally Cartesian as Descartes himself, without the certainty. But if perception is understood as a disclosing of what reality is and how it functions, a very different picture begins to emerge. Perception involves meaning, it is part of a unified activity in which the perceiver attempts to make sense of the world and its structures. This, then, is not a bridge but an interaction in which a being already in the world attempts to make sense of its environment. The position in which to judge a flower as a

flower is not so slippery as to prove that either A) no such privileged position exists therefore all perception is appearance, nor B) no such privileged position exists therefore size, shape and colour are wholly mind-dependent and the outside world is an unjustifiable inference. Rather, the position to judge a flower as a flower is that which enables *flower* to be disclosed qua flower, for the ontological reality of the thing to become present in the awareness of the perceiver.

Before concluding, certain observations are worth making. While it is true that Russell mentions multiple senses, discussions on perception tend to focus around and privilege the sense of sight, especially when advocating for a form of representationalism, and ultimately Russell himself is no exception here. The appearance versus reality problem as instantiated by Ayer's stick and Russell's table are both settled in ocular terms. Ayer says little of the sense-data we would presumably experience if we were to touch the stick in order to ascertain straightness or otherwise, but if sense-data exist they must exist for all senses. While it's true that 'feeling' a rectangle is more of a task than feeling the straightness of a stick in water, the fact that one sense (touch) contradicts the look of a thing given by another sense (sight), points to an important fact, namely that sense perception is multi-modal, and further reinforces the argument that sight perceiving refraction is not a moment where human perception breaks down and appearances impose themselves between perceiver and perceived, but rather further proof that we both accurately track predictable distortions and that reality itself is disclosive, revealing to us via sensory perception the structures of the world and how its laws function. Perceptual variation arguments tend to focus, and take themselves as thereby settled, on issues of sight. Touching the bent stick and realising it is straight is a disclosive moment in which reality teaches us about itself, in particular how light functions in a given context. This highlights the importance of understanding perception as a unitary activity, with each sense contributing to the other, as Aristotle recognised¹⁴. Taken in isolation, the senses are more susceptible to sceptical worries and attacks; it is indicative of a kind of contemporary fragmentation. Perception as an integrated, embodied activity may be an aspect of a more authentic engagement with reality, requiring a more primary and holistic understanding of what the human is, but that is an area for future study.

In conclusion, what are often taken to be variations and illusions should be understood instead as lawful structures, the world behaving consistently under changing relations. Russell's conclusion may appear plausible until we see that he has smuggled into the framework an assumption which we need not accept, nor indeed one that has any claim to validity, i.e., that truthful perception must be invariant. By understanding 'vantage point' in relation to an ontological disclosure relative to the situation of the perceiver, it becomes clear that 'correct' is something mythical which forces us into an unacceptable, sceptical position and invites us to adopt an appearance versus reality dichotomy. This move can be resisted without attempting to deny multiple aspects to a given instance of perception, in recognising the notion that reality is mind-independent, while the abstract

¹⁴ Aristotle, *De Anima*, Penguin, II.6 p172

predicates through which we articulate some of its stable features are communally articulated, i.e., based in agreements, such that geometric and metric predicates are a dimension of our specifically human way of communicating. 'Table' and 'rectangle' are linguistic tools, fixed through shared practice, which enable comprehension, engagement with and manipulation of the shared space of the physical world. Such language does not create the world but stabilises it and communicates intelligible features of mind-independent beings. The intellect does not rescue humans from their faulty perceptions; in fact, the intellect (judgement) is where error is produced, but when judgement is accurate it deepens, integrates and articulates what perception already gives. The world is not dependent on a shared human schema but is navigated by it, and its lawfulness does not reveal a breakdown in human perception (e.g., refraction) and thereby imply a separation of subject from object, but is rather a continual instance and possibility of the world's disclosing itself to us. The so-called problem of illusion dissolves once it is understood that such instances are themselves moments of intelligibility embedded in and concordant with reality's deepest structures, structures which an embodied being open to disclosure is capable of receiving. We might say that 'Illusion' is explained inside perception, not as evidence of its fallibility.

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